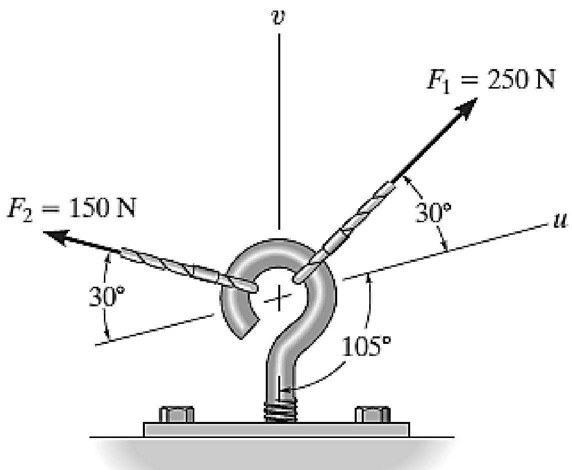
SITUATION: Resolve F1 and F2 into components along the *u* and *v* axes and determine the magnitudes of these components.



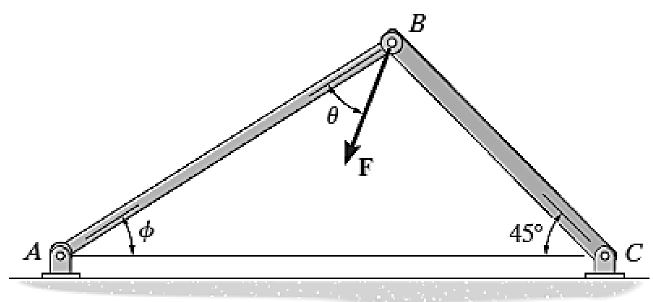
1. F1*v*
A. 183 B. 129 C. 77.6 D. 150

2. F1*u*
A. 183 B. 129 C. 77.6 D. 150

3. F2*v*
A. 183 B. 129 C. 77.6 D. 150

4. F2*u*
A. 183 B. 129 C. 77.6 D. 150

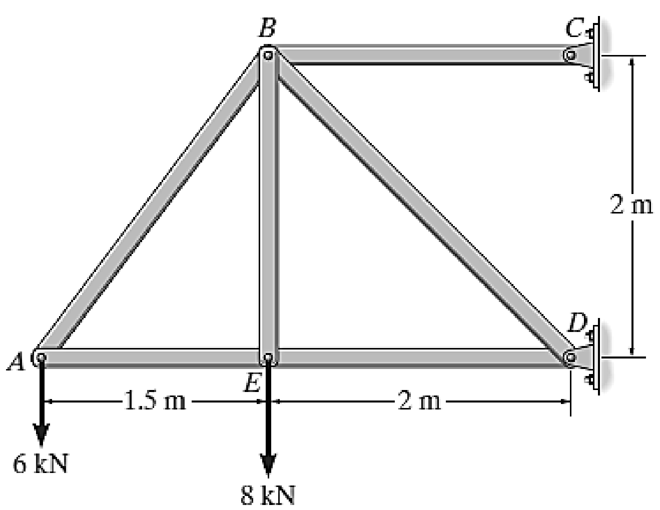
SITUATION. Force **F** acts on the frame such that its component acting along member AB is 650 lb, directed from B towards A, and the component acting along member BC is 500 lb, directed from B towards C. Determine the magnitude of **F** and its direction θ. Set φ = 60°.



5. F.
A. 619 B. 712 C. 815 D. 917

6. θ.
A. 31.8 B. 27.4 C. 22.2 D. 36.6

SITUATION. Determine the force in each member of the truss and state if the members are in tension or compression.

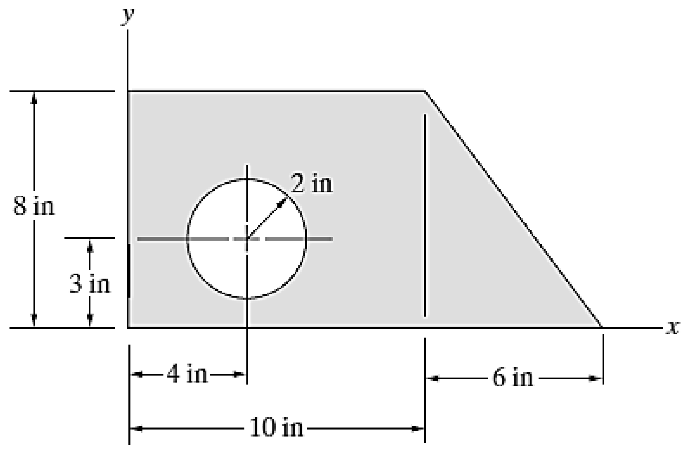


7. F_{AE}.
A. 7.5 (T) B. 4.5 (C) C. 18.5 (T) D. 19.8 (C)

8. F_{BD}.
A. 7.5 (T) B. 4.5 (C) C. 18.5 (T) D. 19.8 (C)

9. F_{BC}.
A. 7.5 (T) B. 4.5 (C) C. 18.5 (T) D. 19.8 (C)

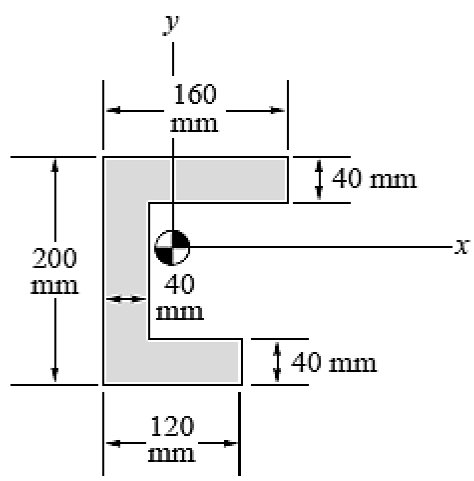
SITUATION. Determine the coordinates of the centroids.



10. $\bar{x}$.
A. 6.97 B. 7.79 C. 8.97 D. 9.79

11. $\bar{y}$.
A. 6.97 B. 5.79 C. 4.97 D. 3.79

SITUATION. From the figure. Determine I_{xc} and I_{yc}.



12. I_{xc} (1 x 10[7]).
A. 7.59 B. 7.69 C. 7.79 D. 7.89

13. I_{yc} (1 x 10[7]).
A. 2.6 B. 3.0 C. 3.4 D. 3.7

SITUATION. Starting from rest, a particle moving in a straight line has an acceleration of *a* = (2*t* - 6) m/s2, where *t* is in seconds. What is the particle's velocity when *t* = 6 s, and what is its position when *t* = 11 s?

14. Velocity (v).
A. 0 B. 2 C. 2.25 D. 3

15. Position (s).
A. 80.7 B. 72.8 C. 68.3 D. 64.9

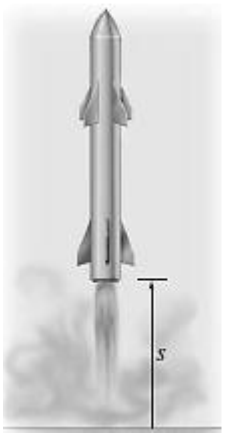
SITUATION. A particle travels along a straight line with a velocity v = (12 - 3*t*2) m/s, where *t* is in seconds. When *t* = 1 s, the particle is located 10 m to the left of the origin. Determine the acceleration when *t* = 4 s, the displacement from *t* = 0 to *t* = 10 s, and the distance the particle travels during this time period.

16. acceleration (a).
A. -28 B. -24 C. -20 D. -31

17. displacement (Δs).
A. -901 B. -880 C. -861 D. -840

18. distance (s).
A. 896 B. 902 C. 912 D. 918

SITUATION. The acceleration of a rocket traveling upward is given by a = 16 + 0.02s2 m/s2, where *s* is in meters. Determine the time needed for the rocket to reach an altitude of s = 100 m. Initially, v = 0 and s = 0 when t = 0.



19. time (t).
A. 4.25 B. 4.86 C. 5.38 D. 5.62

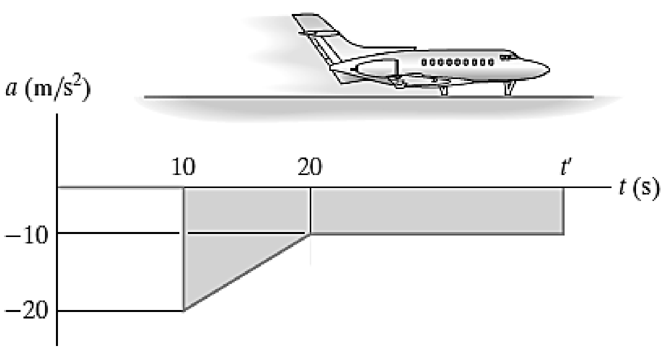
SITUATION. Two rockets start from rest at the same elevation. Rocket *A* accelerates vertically at 20 m/s2 for 12 s and then maintains a constant speed. Rocket *B* accelerates at 15 m/s2 until reaching a constant speed of 150 m/s. Construct the *a*–*t*, v–*t*, and *s*–*t* graphs for each rocket until *t* = 20 s.

20. What is the distance (km) between the rockets when *t* = 20 s?
A. 1.11 B. 2.25 C. 3.36 D. 2.22

21. What is the velocity of rocket A when t = 12 s?
A. 150 B. 195 C. 214 D. 240

22. What is the distance traveled by rocket B when t = 10 s?
A. 750 B. 1440 C. 1110 D. 930

SITUATION. The motion of a jet plane just after landing on a runway is described by the *a*–*t* graph. Determine the time *t*′ when the jet plane stops. Construct the v–*t* and *s*–*t* graphs for the motion. Here *s* = 0, and v = 300 ft/s when *t* = 0.

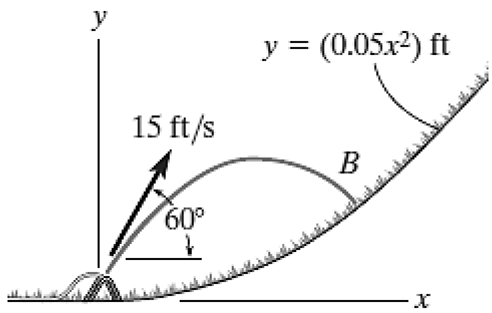


23. time (t') when the jet plane stops.
A. 30 B. 32.5 C. 35 D. 37.5

24. velocity of the jet plane when t = 20 s?
A. 150 B. 125 C. 120 D. 175

25. the displacement between t = 10 s to t = 20 s.
A. 2133 B. 2167 C. 2185 D. 2192

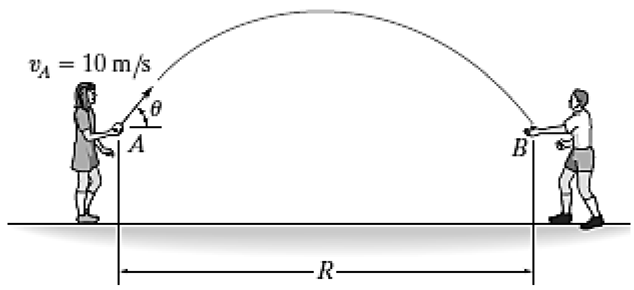
SITUATION. The water sprinkler, positioned at the base of a hill, releases a stream of water with a velocity of 15 ft/s as shown. Determine the point *B(x, y)* where the water strikes the ground on the hill. Assume that the hill is defined by the equation $y = (0.05x^2)$ ft and neglect the size of the sprinkler.



26. point x.
A. 4.15 B. 5.15 C. 6.15 D. 7.15

27. point y.
A. 0.67 B. 1.33 C. 1.67 D. 2.33

SITUATION. The girl at *A* can throw a ball at v*A* = 10 m/s. Calculate the maximum possible range *R* = *R*max and the associated angle *θ* at which it should be thrown. Assume the ball is caught at *B* at the same elevation from which it is thrown.



28. angle *θ*.
A. 58 B. 51.5 C. 45 D. 40

29. Rmax.
A. 10.19 B. 7.29 C. 8.19 D. 9.29

30. Maximum height the ball traveled.
A. 1.85 B. 1.55 C. 2.85 D. 2.55